Day-18
Free CCNA - Part (3)
VLANS

~~Dec~~
oct 24

Contents.

- Native VLAN on a router
- WireShark Analysis (Note bandaina yesko)
- Layer 3 Switching / Multilayer Switching.

Native VLAN on a router.

The best practice is to set native VLAN to unused VLAN. If we want to use native VLAN feature, lets see how can we do it.

Benefit of Native VLAN feature is that the frames in Native VLAN aren't tagged, it's more efficient, each frame ~~is~~ is smaller, so it allows device to send more frames per second. The network int. is that of day 17.

There are 2 methods of configuring native VLAN on router

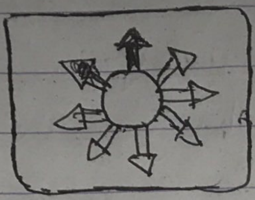
i) Using command "encapsulation dot1q vlan-id native" on router sub-interface.

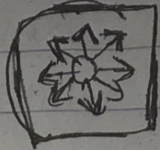
```
R1 # int g0/0.10
R1 # encapsulation dot1q 10 native
R1 # ip add 192.168.1.0 255.255.255.192 {network add?
```

ii) Configure ip add. for native VLAN on router's physical interface (we don't need encapsulation dot1q vland id commnd).

```
R1 # no interface g0/0.10  (deletes subinterface)
R1 # int g0/0
R1 # ip add 192.168.1.62 255.255.255.192
R1 #
```

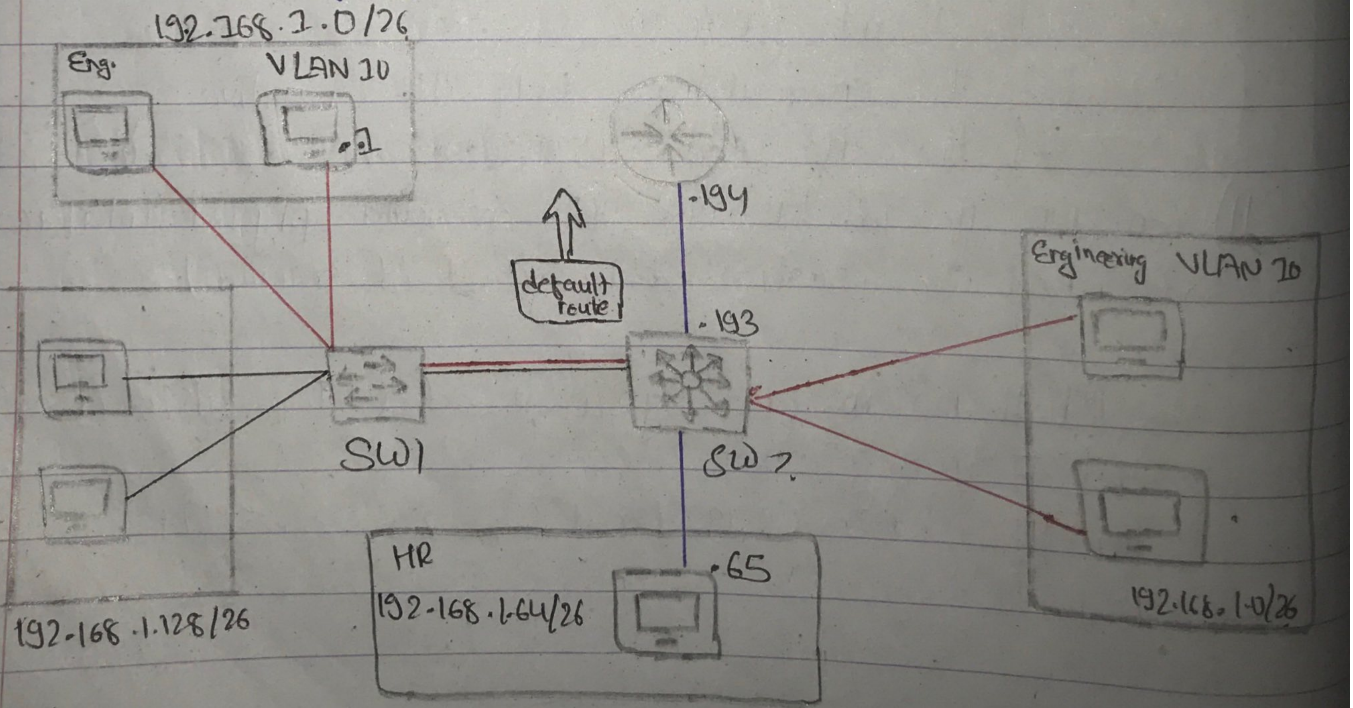
Layer 3 switch/Multilayer switch,





- A multilayer switch is capable of both switching & routing.
- It is 'layer 3 aware'.
- We can assign IP addresses to its interfaces, like a router
- You can create virtual interfaces for each VLAN, and assign IP addresses to those interfaces.
- You can configure routes on it, just like a router.
- It can be used for inter-VLAN routing.

Network topography.


192.168.1.0/26
Eng.
VLAN 10
.1
.194
default route
.193
Engineering VLAN 10
SW1
SW2
HR
.65
192.168.1.64/26
192.168.1.128/26
192.168.1.0/26

SVI(Switch-Virtual Interfaces)

- SVI are the virtual interface we can assign ip addresses to in a multilayer switch.

- Configure each PC to use the SVI(NOT the router) as their gateway address.
- To send traffic to different subnets/VLANS, the PCs will send traffic to switch, and the switch will route the traffic.

Inta-VLAN routing via SVI

Router Configuration

a) If we have already made subinterfaces remove it

`no interface g0/0.10`

Then reset g0/0 to default settings.

b) `default interface g0/0`

c) Then, 'int g0/0'

d). `ip add 192.168.1.194 255.255.255.252`

Switch Configuration.

i) Reset g0/1 interface to default settings

`default interface g0/1`

ii). Enable layer 3 routing on switch.

`ip routing.`

iii) `int g0/1`

iv) Change int from layer 2 switched port to layer 3 routed port.

`no switchport.`

v) `ip add 192.168.1.193 255.255.255.252`

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vi) ip route 0.0.0.0 0.0.0.0 192.168.1.194
(default route pointing to R1)

vii) Then to confirm use
Show interfaces status.
(Gi0/2 connected route auto auto unknown

viii) int vlan10
ix) # ip add 192.168.1.62 255.255.255.192
x) # no shutdown.
xi) do sh ip int brief. (vlan must be in up/up).
(SVIs are shutdown by default, so remember to use no shutdown to enable them)

Requirements for SVI to be up/up

i) The ~~vlan~~ VLAN must exist on switch.
{switchport access vlan 10} yesto or trunk mode use garera vlan ar jst vlan 10 garera bhye ni VLAN banakohunuparxa.

ii) The switch must have at least one access port in the VLAN in an up/up state AND/or one trunk ~~post~~ port that allows the VLAN that is in up/up state.

iii) The VLAN mustn't be shutdown (use no shutdown command

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